Interview Questions & Answers — Introduction to Load Balancing

1. What problem does load balancing solve?

Answer:
Load balancing solves the problem of handling increasing traffic in a scalable and reliable manner. As the number of users grows, a single server can become overwhelmed by incoming requests. A load balancer distributes traffic across multiple servers, ensuring that no single server carries all the workload. This improves system performance, enables scalability, and helps maintain availability even when individual servers experience issues.

2. Why does a single server eventually become a bottleneck?

Answer:
Every server has finite resources, including CPU, memory, storage, and network bandwidth. As user traffic increases, these resources become increasingly utilized until the server reaches its capacity limits. Once this happens, response times increase, requests begin to queue, and users may experience timeouts or failures. Regardless of how powerful a server is, there is always a maximum capacity it can handle, making it a bottleneck as the application grows.

3. What is the difference between vertical scaling and horizontal scaling?

Answer:
Vertical scaling involves increasing the resources of an existing server, such as adding more CPU cores, memory, or storage. Horizontal scaling involves adding more servers and distributing the workload across them.

Vertical scaling is generally simpler because the application remains on a single machine. However, it has physical and financial limits. Horizontal scaling requires additional infrastructure and traffic distribution mechanisms but provides significantly greater scalability and fault tolerance, making it the preferred approach for large-scale systems.

4. Why is horizontal scaling often preferred over vertical scaling?

Answer:
Horizontal scaling is often preferred because it provides a more sustainable path for long-term growth. While vertical scaling can temporarily increase capacity, there is always a limit to how much a single server can be upgraded. Additionally, larger servers become increasingly expensive and create a greater dependency on a single machine.

By adding more servers, organizations can continue increasing capacity as demand grows. Horizontal scaling also improves fault tolerance because the system can continue operating even if one server fails.

5. What challenge arises when an application is deployed across multiple servers?

Answer:
Once multiple application servers are introduced, incoming requests must be distributed among them efficiently. Users should not need to know which server to connect to, and traffic should be directed in a way that makes effective use of available resources.

Without a mechanism to manage this distribution, some servers may become overloaded while others remain underutilized. This challenge creates the need for a load balancer to act as a central traffic distribution layer.

6. What role does a load balancer play in a distributed system?

Answer:
A load balancer acts as a single entry point for incoming requests. Instead of clients connecting directly to individual application servers, they send requests to the load balancer, which then forwards those requests to one of the available backend servers.

This abstraction simplifies client interactions, enables horizontal scaling, and allows the system to distribute traffic efficiently. It also provides a foundation for improving availability and reliability within distributed architectures.

7. How does load balancing improve scalability?

Answer:
Load balancing improves scalability by allowing traffic to be distributed across multiple servers rather than relying on a single machine. As traffic increases, additional servers can be added to the system, and the load balancer can begin routing requests to them.

This approach enables capacity to grow incrementally and helps organizations handle increasing workloads without requiring major architectural changes or constant hardware upgrades.

8. How does load balancing improve availability?

Answer:
Load balancing improves availability by reducing dependency on any single server. When multiple servers are available to handle requests, the system can continue operating even if one server becomes unavailable.

This redundancy helps minimize downtime and ensures that users can continue accessing the application despite infrastructure failures, maintenance activities, or unexpected outages.

9. What happens if one server fails in a load-balanced architecture?

Answer:
In a load-balanced architecture, a failed server can be removed from active traffic routing, allowing requests to be directed to the remaining healthy servers. As a result, the application continues serving users with minimal disruption.

This capability significantly improves reliability because a single server failure no longer causes the entire application to become unavailable. The impact of failures is isolated, making the system more resilient.

10. Why is load balancing considered a foundational component of modern system design?

Answer:
Load balancing is considered foundational because it enables many of the characteristics expected from modern applications, including scalability, high availability, reliability, and fault tolerance.

As systems grow beyond a single server, some mechanism is needed to distribute traffic and manage multiple instances. Load balancing provides this capability and serves as a critical building block for distributed systems, cloud-native architectures, and large-scale applications.

11. When should you introduce a load balancer into a system?

Answer:
A load balancer should be introduced when a single server can no longer meet performance, scalability, or availability requirements. This typically occurs when traffic increases significantly, when multiple application servers are required, or when the business demands higher uptime and reliability.

Introducing a load balancer becomes especially valuable when organizations begin adopting horizontal scaling strategies and need a reliable way to distribute traffic across multiple servers.

12. Walk me through the evolution from a single-server application to a load-balanced architecture.

Answer:
Most applications begin with a single server because it is simple and cost-effective. As traffic grows, that server eventually reaches its capacity limits, leading to performance issues and reduced reliability.

The first response is often vertical scaling by upgrading the server's resources. However, this approach eventually reaches practical limits. To continue growing, additional application servers are introduced through horizontal scaling.

Once multiple servers exist, traffic must be distributed among them efficiently. A load balancer is added in front of the servers to act as a single entry point and route requests appropriately. This evolution enables the system to handle larger workloads, improve availability, and become more resilient to failures.